AI CLUSTERS

From Single Data Centers to Multi-Region AI Clusters: The Rise of Scale Across

Spencer Giacalone *Senior Director Cloud AI Architecture, Ciena*

Helen Xenos *Senior Director Product Marketing, Ciena*

OCP BRONZE™ Member

Optical Is The Foundation For Scale Across

Training traffic at distance: large, bursty, loss- and delay-intolerant

Congestion Control Challenges
- Incast, protocol under/reaction
- Buffer crediting, host buffers, replay optimizations
- Lossless protocols (UEC, MRC, MetaRoCE, etc)

Model Optimizations
- Optimized (e.g., hierarchical) collectives
- Optimized multi-dimensional parallelism
- Overlapped comms / compute (hide by compute)

Host and protocol fixes manage contention, not required capacity

Optical capacity is the foundation for Scale-Across

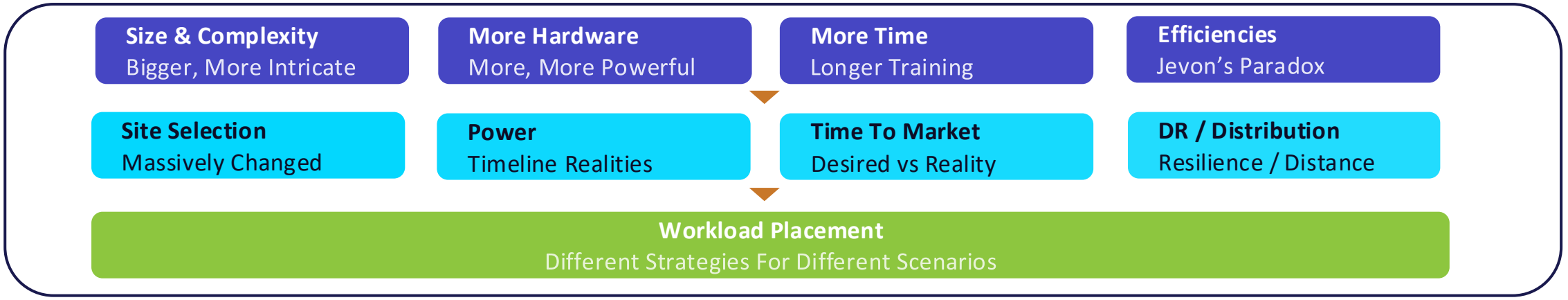
Blast radius reduction

Per flow / packet drop probability

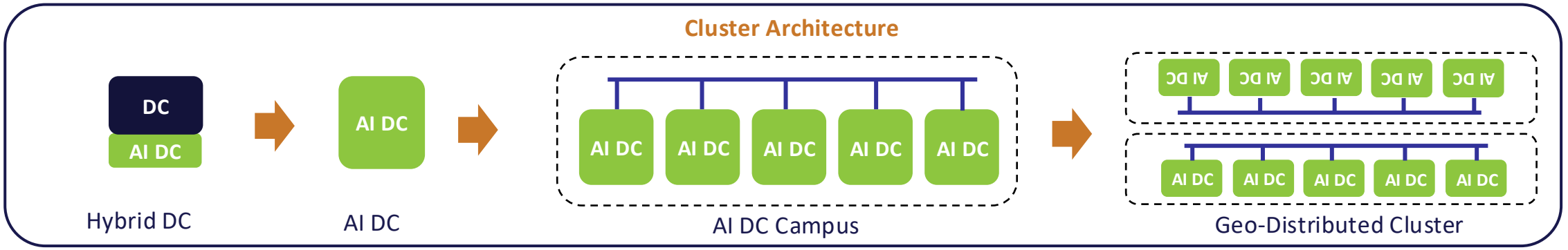
Lower contention

Headroom, load balancing burst absorption

Getting to Scale Across



"...as gigawatt-scale energy demands and grid constraints prevent single locations from supporting hundreds of thousands of synchronous GPUs... ...we are compelled to expand to multiple buildings across distributed regions. ...inter-building distances are dictated by power, grid, and cooling constraints"[1]



1) "ScaleAcross Explorer: Exploring Communication Optimization for Scale-Across AI Model Training", li, Golden, Gangidi, et al, Harvard University, Meta Platforms, 2026

Scale Across Defined

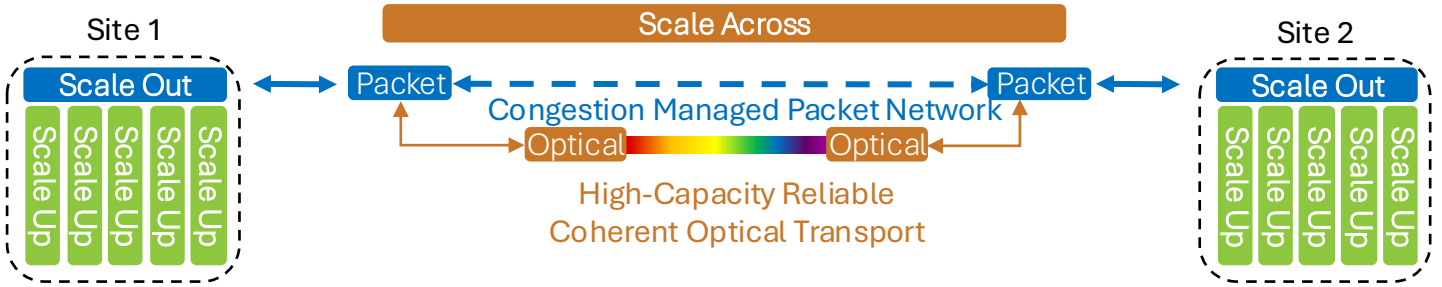
Extending lossless connectivity[1] between datacenter campuses or regions to support the training of single AI model

1. *Packet layer: Congestion management, fairness, load balancing at the IP layer & above*
2. *Optical layer: A dedicated **coherent optical network** with enough capacity to support large bursty loss sensitive workloads*
   - *Scaled Coherent DWDM is a main differentiator between Scale Out and Scale Across*

Scale Across:

Extending lossless connectivity between datacenter campuses or regions to support the training of single AI model

Packet layer:
Congestion & load balancing



Optical layer:
Extreme coherent capacity

What Is Scale Across Used For, Anyway?

Pipeline Parallel[1]

HSDP/FT-HSDP[2]

1) "ScaleAcross Explorer: Exploring Communication Optimization for Scale-Across AI Model Training", li, Golden, Gangidi, et al, Harvard University, Meta Platforms, 2026
2) "Training LLMs with Fault Tolerant HSDP on 100,000 GPUs", Salpekar, Varma, Yu, Gangidi, et, al, Mata Platforms, Thinking Machine Labs, The Ohio State University, Genesis Molecular AI, 2026

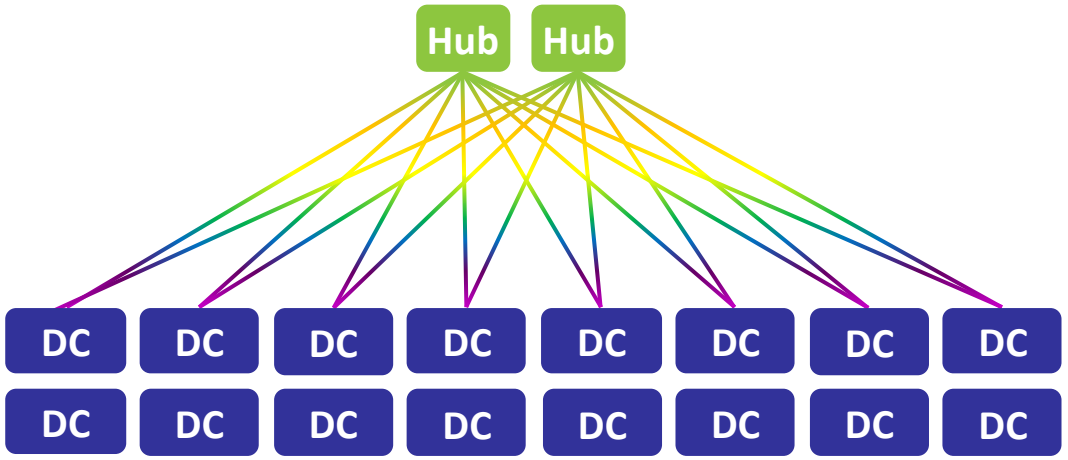
IMDD vs Coherent

Goal: Drive maximal campus capacity over distance, through a constrained set of optical fibers

Distance	**O-band IMDD (Campus)** • **Up to ~10 km** (DR /FR/LR)	**C+L coherent DWDM (Inter-Campus)** • **~120 km** per span, **1,000 km+** with ILAs
Capacity per fiber	**Single Tb/s** • Amplitude modulation w/ direct detect • 2 bits/symbol, coarse channels	**Tens of Tb/s** • Amplitude + phase modulation + dual polarization • **8 bits/symbol (DP 16QAM) + WDM + C+L bands**
Impact	**Linear fiber growth** • **One port** per fiber	**Scaled fiber growth** • **Many router ports** per fiber

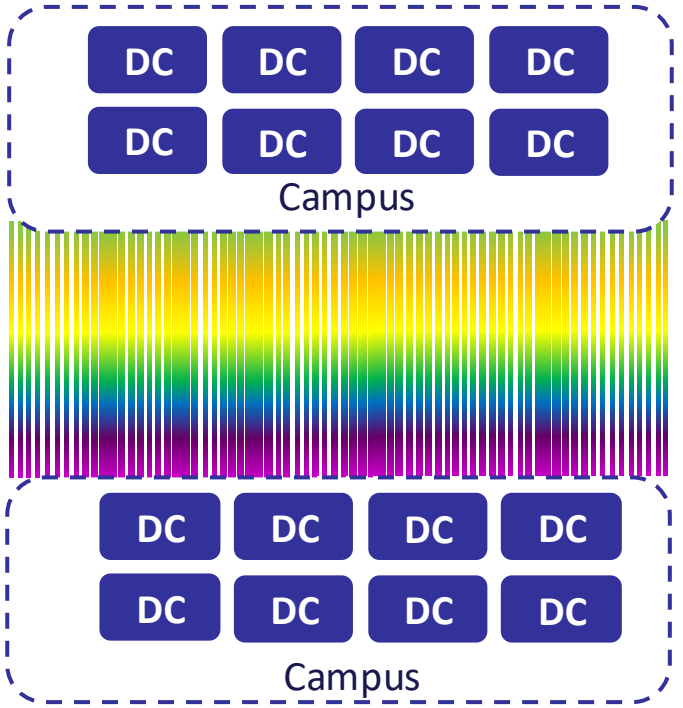
Coherent optics cost more, limits campus use. Coherent-Lite closes gap

Scale Across Capacity



Front End Region- Circa 2020

14x



Distributed AI Cluster

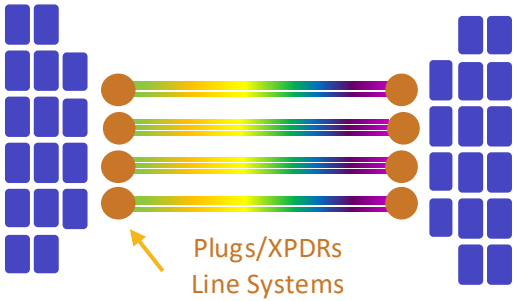
What is the oversubscription of Scale Across?

>8:1[1]

1) "ScaleAcross Explorer: Exploring Communication Optimization for Scale-Across AI Model Training", li, Golden, Gangidi, et al, Harvard University, Meta Platforms, 2026

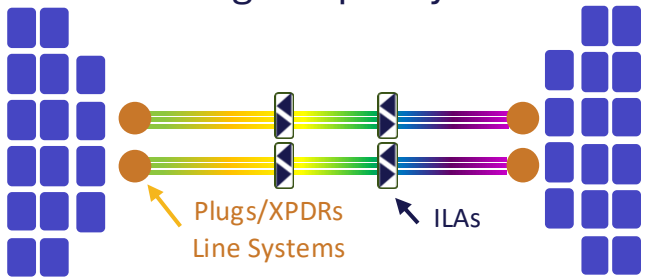
Scale Across Designs

Regional
“Near Scale Across”
-Extreme Capacity-



- <150km- Extended campus
- **Extreme** capacity
- **Plugs/XPDRs, line systems- No ILAs**
- **P2P, C+L, full fiber Increments**

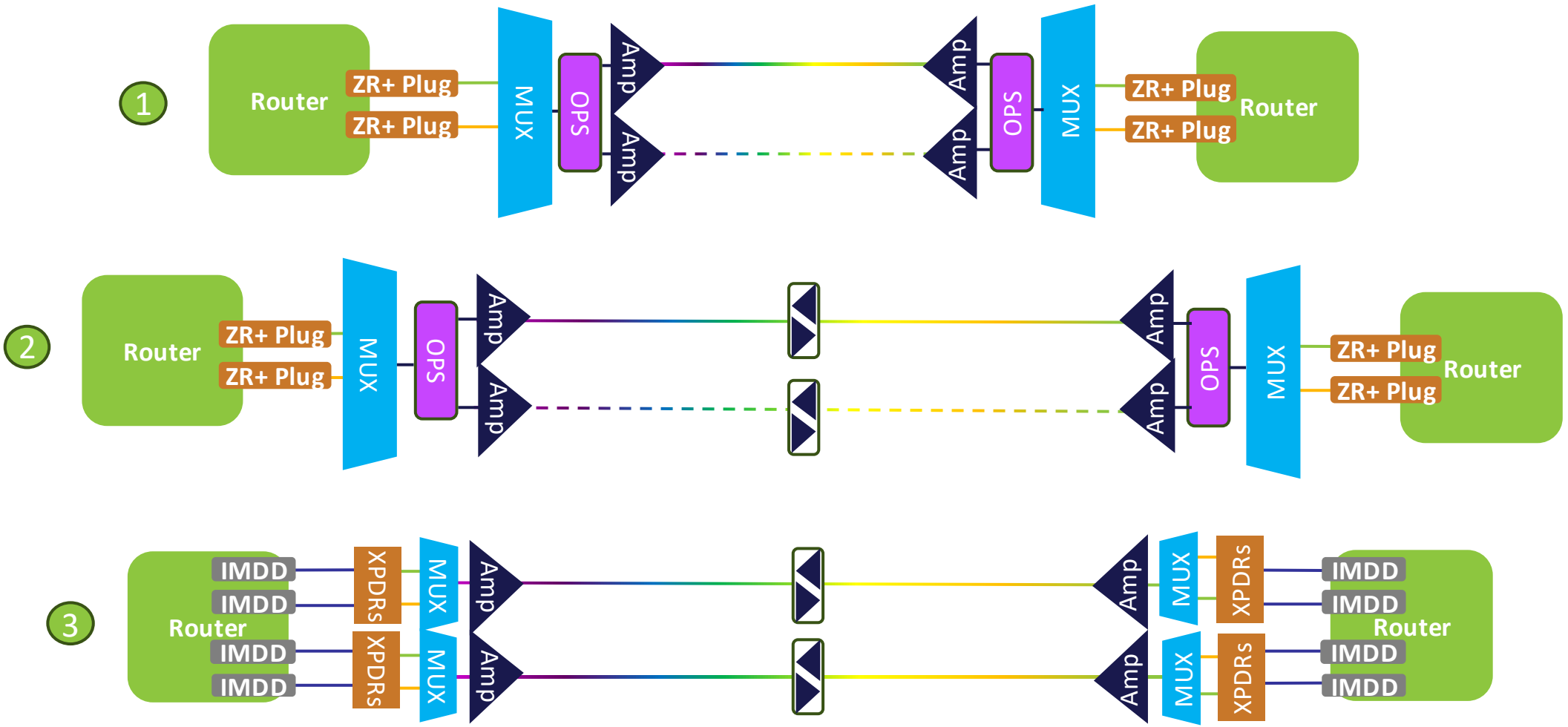
Inter-Regional
“Far Scale Across”
-High Capacity-



- >150km- Truly distributed
- High capacity, but not as extreme
- **Plugs/XPDRs, line systems & ILAs**
- **P2P, C+L, full fiber Increments**

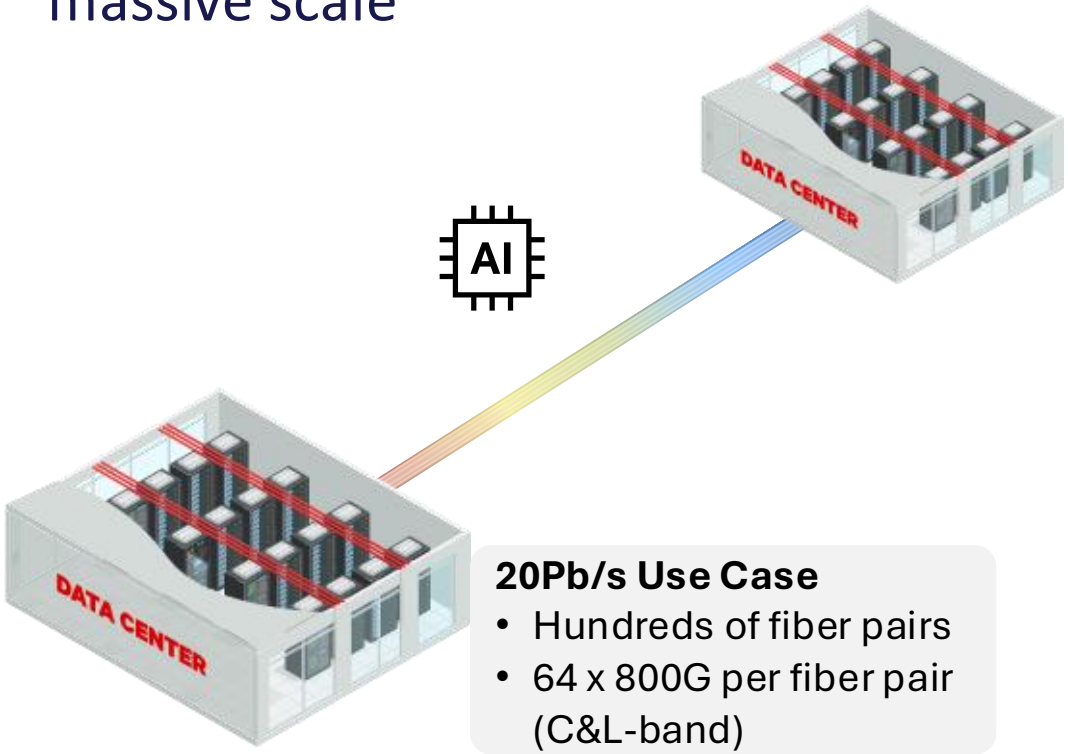
Fiber Is The Unit of BW
Full Fill C+L

Implementation Trends



Scale Across networking has arrived

Optimized for simplicity, density, and massive scale



20Pb/s Use Case
- Hundreds of fiber pairs
- 64 x 800G per fiber pair (C&L-band)

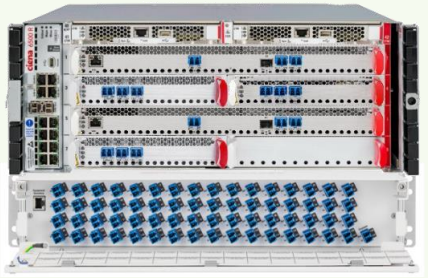
WaveLogic coherent optics



- WaveLogic coherent pluggable transceivers
- Higher performance coherent optics, including full spectrum transponder in future

RLS modern line system

RLS C&L terminal config



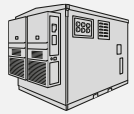
RLS Hyper-Rail



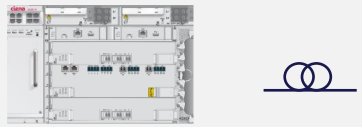
16 x C- and L-band fiber pairs in compact chassis

Redefining photonic infrastructure for AI scale

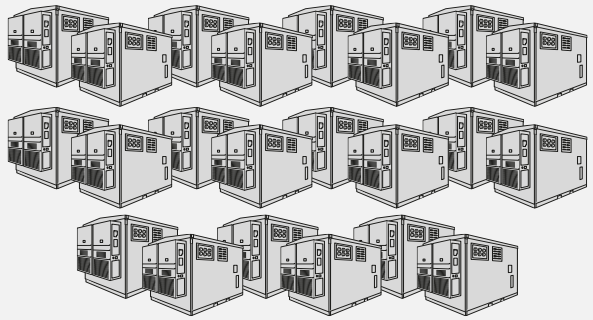
Line amp sites TODAY
4 rails/rack, 16 rails/hut
All capacity within one hut



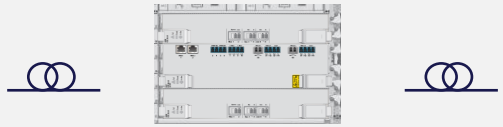
4 rails / rack
1 hut



20 Pb/s AI Network
Using conventional line system designs → 22 huts required



4 rails / rack
22 hut



Multi-rail photonic architecture

32x
density improvement

RLS Hyper-Rail:
20 Petabit/sec
1 hut



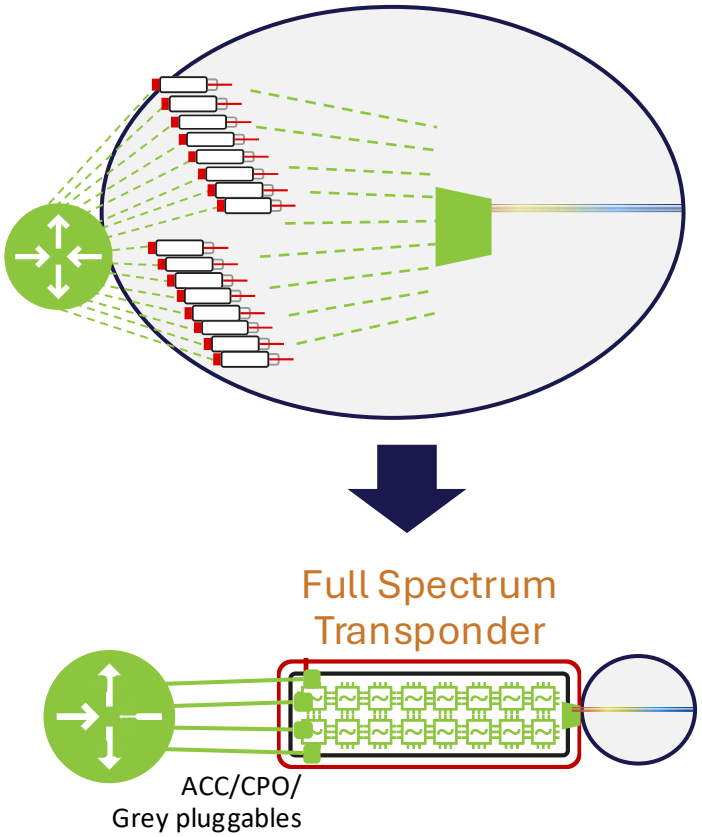
128 rails / rack
1 hut



Multi-rail photonic architectures enable AI-scale capacity with a fraction of the infrastructure

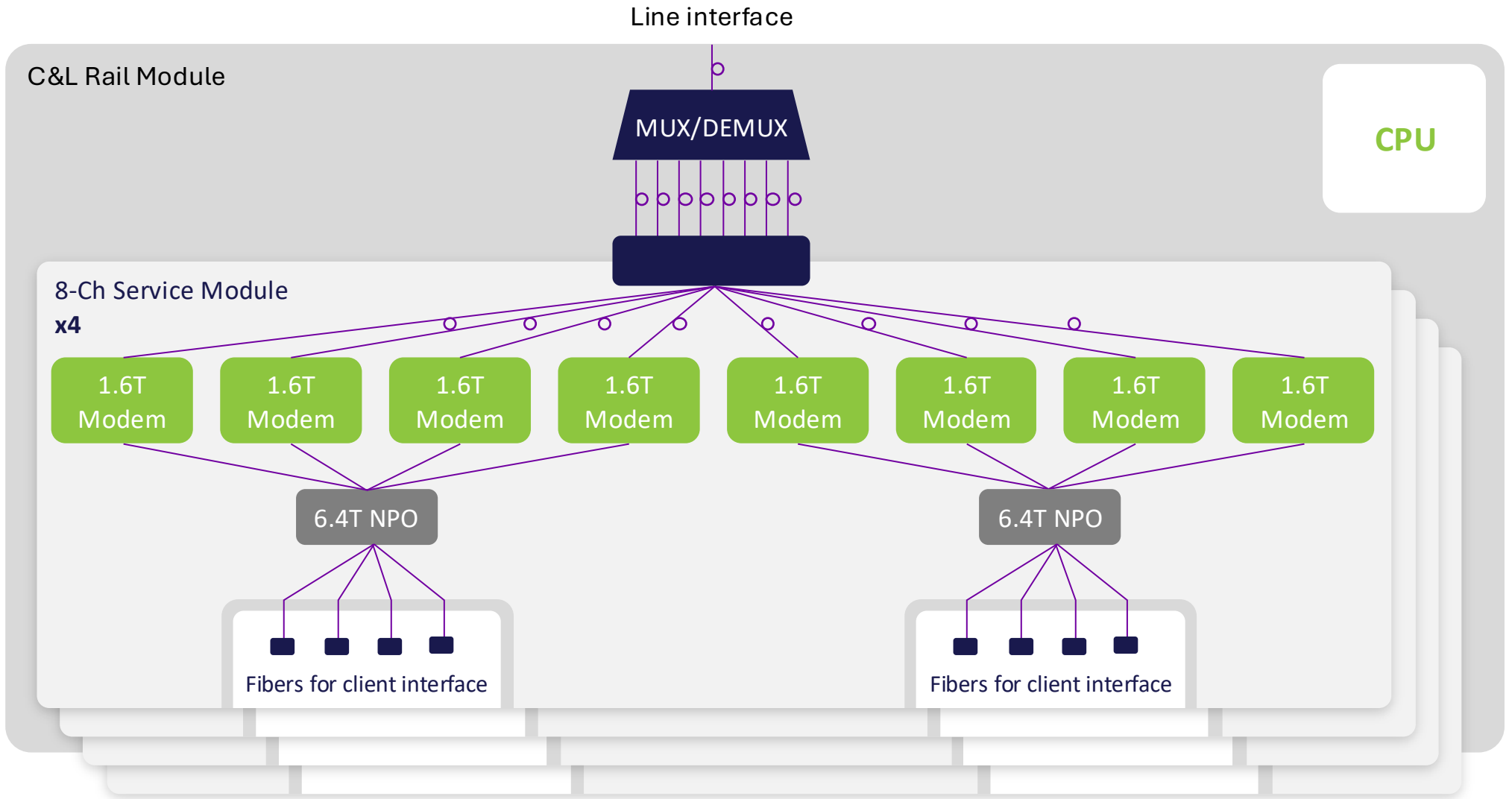
At scale, the fiber becomes the unit of deployment

- Today: wavelength-by-wavelength deployment
  - 128 x 400G x 2 pluggables per fiber pair
  - 64 x 800G x 2 pluggables per fiber pair
- Scaling this model for 10s-100s fiber deployments becomes operationally complex.
- **Full spectrum transponder**
  - Lights the entire fiber at once
  - Returns to grey optics and external transponders
  - Reduces reliance on DWDM expertise
  - Enables cookie-cutter deployments
  - Can be higher performance without pluggable power constraints

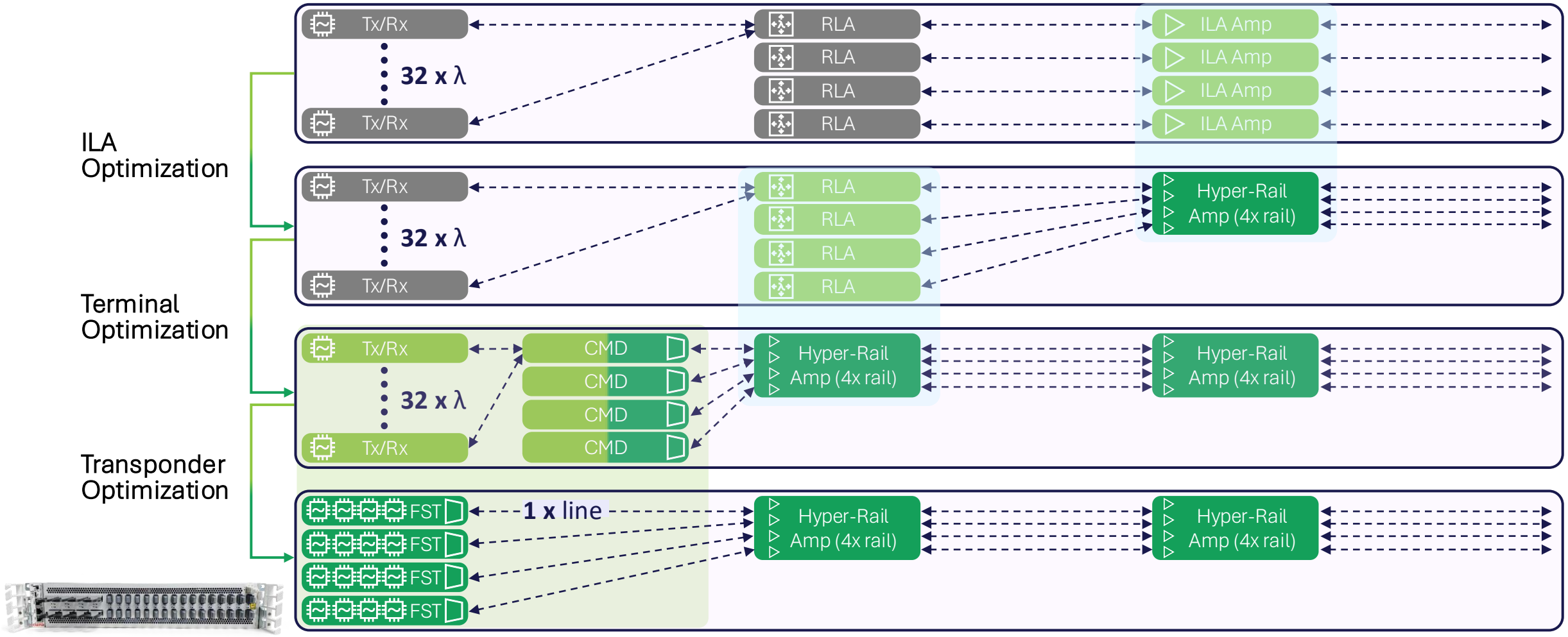


Optimizing for deployment velocity increasingly favors full-fiber architectures

What a 51.2T C&L full-spectrum transponder could look like



Architecture Evolution toward Full-Fiber Architectures



Enables repeatable, cookie-cutter deployments at AI scale

Scale Across Trends for 2027 (And Beyond…)

Today	Tomorrow
Near Scale Across • Near Scale Across = No ILA planning, cost	**Customers Go Far Scale Across** • ILA Planning, co-development • Capacity increasing to near scale across levels
Simplest, Fastest Deployment • Plugs + Fixed Filters + Raman	**More Diverging, More Flexible Designs** • Plugs vs FST • Raman vs EDFA vs Both • WSS vs fixed
Dedicated Scale Across Deployments • Training Only Routes	**The Optical AI Backbone Emerges** • Same equipment, same network, AI team leads • Inference, training, front end
Optical As An Enabler • Optical path protection to save router ports	**Optically Oriented Network Design** • Regional, DCI, and even in DC (OCS)

*Fixed filter Raman fastest delivery, lowest power that needed no hut

Call To Action- Lossless Physical Layer

Problem To Solve

Ambient bit errors- SNR → Coherent FEC designed for loss tolerant applications → Scale Across link speed, radix, application changes requirement

Lossless Transport Deals With Via Retransmissions

Better fiber (already done) → Modem power (FEC/Baud)- FEC provability → Limit distance (not desired)

Solution
Provide a provably lossless physical layer (e.g., 1e-24 post FEC BER)

How To Get Involved
Email: jharley@ciena.com, spencer@ciena.com

Let's start tracking Scale Across in the AI Clusters group
https://www.opencompute.org/w/index.php?title=OCP_Open-Cluster-Designs-for-AI

Takeaways

- **Scale Across extends lossless connectivity between datacenter campuses or regions to support the training of single AI model**
  - Coherent networks are required for Scale Across
  - New platforms being created: multi-rail photonics, full spectrum transponder

- **OCP possibilities:**
  1. Contiguous campus definition
  2. Standardize on scale across definition
  3. Standardize on FEC requirement for lossless connectivity

- **Inference:**
  - Large workload, more distributed; many of the same optical building blocks


Thank
You!
OCP
GLOBAL
SUMMIT
October 12-15, 2026
San Jose, California